Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/17/2017
PI Name:	Hoffman, Larry Ph.D.		
Project Title:	Gravity-Induced Plasticity in Mammalian Utricular Hair Cells: Intrinsic or Multisensory?		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	90095-1624	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	03/31/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2019 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: Extended to 8/31/2018 per F. Hernandez/ARC (Ed., 7/13/17) NOTE: Extended to 7/31/2017, per NSSC information (Ed., 6/15/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schweizer, Felix Ph.D. (University of California, Los Angeles) Fuller, Charles Ph.D. (University of California, Davis)		
Grant/Contract No.:	NNX13AL99G		
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Performance Goal Text:			

Task Description:	This proposal is being submitted to the research topic area Ground based research designed to lead to space flight announced in NASA NRA (NASA Research Announcement) NNH12ZTT001N. We are proposing a series of studies to investigate whether the plasticity in the inner ear utricle is affected by non-vestibular load-sensing physiologic systems. Potential interactions between load-sensing systems might offer cross-modal synergy under altered gravity conditions, e.g., loading of the utricle might directly affect other systems (e.g., immune, osteogenic) that are currently subject to significant countermeasure efforts. In addition, the proposed research endeavors to elucidate how this plasticity is manifested in alterations of the activity of neurons that signal such changes to the central nervous system. We plan to use mouse models that systematically exclude the interaction of particular components of the vestibular load-sensing system. These models represent valuable adjuncts that can be implemented in future flight studies to address countermeasures driven by the recruitment of multiple sensory systems. Furthermore, the present investigation serves as a segue between altered gravity exposure durations that have been focused upon the duration of space shuttle missions and the much longer exposures incumbent with experiments aboard the International Space Station.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research explores the natural capabilities of utricular hair cells to undergo modifications in their synaptic connections with neurons that communicate sensory signals to the central nervous system as a result of exposure to altered gravitational fields. The ambient gravitational field is a fundamental component of the natural sensory environment, hence a principal aspect of our research is to explore the "sensory learning" that may result when changes in gravity are imposed. These fundamental findings are the foundation for future experiments to reveal the cellular mechanisms responsible for these changes. Since synaptic decline may reflect one form of natural vestibular dysfunction due to aging, pathology, or ototoxicity, an understanding of this repertoire and the cellular mechanisms is the first step toward designing rehabilitative strategies. Our research is designed to develop a systematic laboratory protocol in which these natural capabilities can be explored.
Task Progress:	Establishment of a colony of otoferlin-null mice: Otoferlin is a critical protein involved in neurotransmitter exocytosis in mammalian hair cells. A critical component of our experimental strategy is to evaluate the distribution of synapses in utricular hair cells under normogravic and hypergravic conditions. This task has taken several years, due to difficulties in breeding these animals. However, they are now well established and we have a vigorous colony to provide animals for this investigation. Quantitative analyses of synapse distributions in the mouse utricle: We have completed the experimental phase of this task, which included immunohistochemical labeling and confocal imaging of synapses in utricular hair cells in normal and otoferlin-null mice. These data are currently undergoing analysis. The results are crucial to the hypergravity phase of this investigation. Electrophysiologic assessment of exocytotic capacity in mouse utricular hair cells: We are working on revising our approach to these experiemnts in which capacitance change is measured following depolarization in mouse utricular hair cells. We are developing the methods to conduct these experiments in a epithelial slice preparation. Hypergravity-exposure of normal mice: We have completed the preparatory work required for conducting the hypergravity component of this investigation in the laboratory of Charles Fuller (University of California (UC) Davis). We have secured approval for this work from the UC Davis Institutional Animal Care and Use Committee (IACUC). We plan to conduct an initial series in normal mice, which will provide critical background for hypergravity exposure in otoferlin-null and otoconia-deficient mice.
Bibliography Type:	Description: (Last Updated: 10/05/2022)
Articles in Peer-reviewed Journals	Sultemeier DR, Choy KR, Schweizer FE, Hoffman LF. "Spaceflight-induced synaptic modifications within hair cells of the mammalian utricle." Journal of Neurophysiology. 2017 Jun 1;117(6):2163-78. Epub 2017 Feb 22. https://doi.org/10.1152/jn.00240.2016 ; PMID: 28228581 ; PMCID: PMC5454470 , Jun-2017